Carnegie Mellon University

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Mechatronics Design – Spring 2026 – Dance Robot

Design Proposal

Team 2

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1. Project description

This project develops a compact, legged, human-carrying assistive mobility robot for Kinematic/Kinesthetic, a live performance by the AXIS Dance Company. The goal is to transport a wheelchair user without wheels as the primary locomotor, using articulated legs to enable expressive, choreographed motion and zero-radius turning.

The project frames assistive technology as an artistic medium rather than a device meant to "fix" disability. The design is intended to be compact, safe, and stage-ready, and the current concept includes autonomous features. These statements and requirements are derived from the project specifications as put forth by our partnership with AXIS.

2. Design requirements

2.1 Explicit requirements from project specifications [1]

2.1.1 Form and mobility

A. Locomotion must be controlled/initiated through use of legs. Load bearing may be supported by auxiliary wheels, treads, or an alternative mechanism.
B. Footprint shall not exceed 5 ft × 5 ft.
C. User ride height shall be 2 ft to 3 ft off the ground.
D. Device mass shall not exceed 150 lb.
E. Device shall carry a person up to 170 lb and account for additional dynamic forces during use.
F. User may sit cross-legged or in a standard seated chair position.
G. Device must travel in at least two directions: forward and backward.
H. Device must have a zero-degree turning radius, with clockwise and counter-clockwise rotation.
I. Device must reliably navigate to a designated point in space within a tolerance of 6 in within 1 minute.
J. Device must be shippable in a 3 ft × 3 ft × 3 ft case, disassembled into no more than 7 parts.
K. Ability to fit through a 36 in door when folded (no disassembly required) is preferred.

2.1.2 Power and cost

A. Battery life shall be at least 30 minutes.
B. Guideline reimbursable limit for total purchased parts/materials cost is approximately $900.

2.1.3 Construction, aesthetics, and safety

A. Device must be robustly constructed (nuts/bolts, machine screws, cable ties, proper soldering, etc.). No prototype kits or toys may be used. Appearance counts.

B. Device must not damage anything with which it interacts. No rough edges, burrs, exposed screw tips, etc.
C. Ride must be sufficiently smooth for the user to ride safely without being strapped to the device.
D. Device must include emergency stop controls.
E. Surrounding environment safety must be considered (robot is used around people during performance).

2.1.4 Sound-level requirements

A. Sound level shall remain under 70 dB during normal operation (traversal, rotation, etc.) measured at 5 ft from robot center using the Decibel X application.
B. Peaks of 85 dB are acceptable only for short periods; total peak duration shall not exceed 1 s for every 60 s of operation.

2.1.5 Special restrictions

A. Teams may not use a laptop in the final machine (allowed for testing/debug).
B. A cell phone or webcam may be used only if it remains part of the machine throughout the course.

2.1.6 Setup and operation

A. Assembly from packed states shall take no more than 15 minutes and shall not require soldering, heavy machinery, etc.
B. Control options allowed include: pre-programmed path, remote control by an operator 20 ft to 50 ft away, or user joystick control.

2.1.7 Final demonstration tasks and scoring constraints

A. The system must complete taped-course tasks while carrying a payload of 150 lb
    a. Reach a designated point 10 ft away in 1 minute via a straight path.
    a. Reach a designated point 10 ft away in 90 seconds via a curvilinear path.
    b. Reach a point 10 ft away in 90 seconds driving backward via a straight path.
    c. Reach a point 10 ft away in 2 minutes driving backward via a curvilinear path.
B. The system must withstand the impact of a 25 lb weight dropped from a height of 5 ft.
C. Performance scoring includes task completion time bands, penalties for deviating from the designated path, and penalties for manual intervention.

2.2 Implied requirements

A. "Appearance counts" and the "no rough edges" rule imply a need for professional finishing like sanding, painting, or deburring.
B. Since the user is 2'–3' high and unstrapped, the gait must remain extremely level to prevent tipping.

C. A 6" navigation tolerance suggests the need for encoders or sensors rather than simple "timed" motor movements.
D. To stay under 70dB, we must likely avoid noisy metal gears or high-frequency motor whine.
E. The 30-minute battery life and "no benchtop power" rule imply a need for on-board voltage regulation and battery monitoring

2.3 Requirements stemming from autonomy

Camera-driven autonomy is the project's distinguishing feature: perception is used first to enforce safety around the rider and nearby performers, and then to enable reliable point-to-point navigation across the performance space.

A. Goal requirements
    a. Any computer required for perception and autonomy must remain on-board.
    b. The robot has at least one autonomous operating mode in addition to manual control options.
    c. The perception stack produces a continuously updated hazard field (free/occupied or safe/unsafe space) from on-board sensing at a responsive rate while moving.
    d. The robot maintains a buffer around detected obstacles; any violation triggers an automatic stop.
    e. Emergency stop controls that immediately inhibit motion commands and place the robot in a safe state.
    f. The robot can navigate to a designated waypoint within a 6-inch tolerance within 1 minute using only on-board components.
    g. Robot autonomy continues to enable forward/backward travel and zero-radius clockwise/counter-clockwise rotation.
    h. The robot detects unlikely sensing inputs and automatically falls back to safe state.
    i. The robot publishes telemetry sufficient for safe testing and debugging (pose estimate, commanded velocity, safety state, and power status) while allowing manual override at any time.
    j. The power system can sustain autonomous sensing and navigation operation for at least 30 minutes on battery.
B. Stretch goal requirements
    a. The robot detects moving obstacles and estimates their short-horizon motion to reduce nuisance stops in the presence of nearby performers.
    b. The robot autonomously re-routes around moving obstacles while maintaining a minimum separation buffer.
    c. The autonomy stack is resistant to lighting and texture changes typical of performance spaces, maintaining stable perception outputs.
    d. The system shall support choreographed multi-waypoint sequences where timing and path shape are repeatable run-to-run using on-board sensing.
    e. The system provides a "safety bubble" visualization or status indicator that communicates when the robot is in safe-to-approach vs stop/avoid states to nearby assistants/operators.

3. Functional architecture

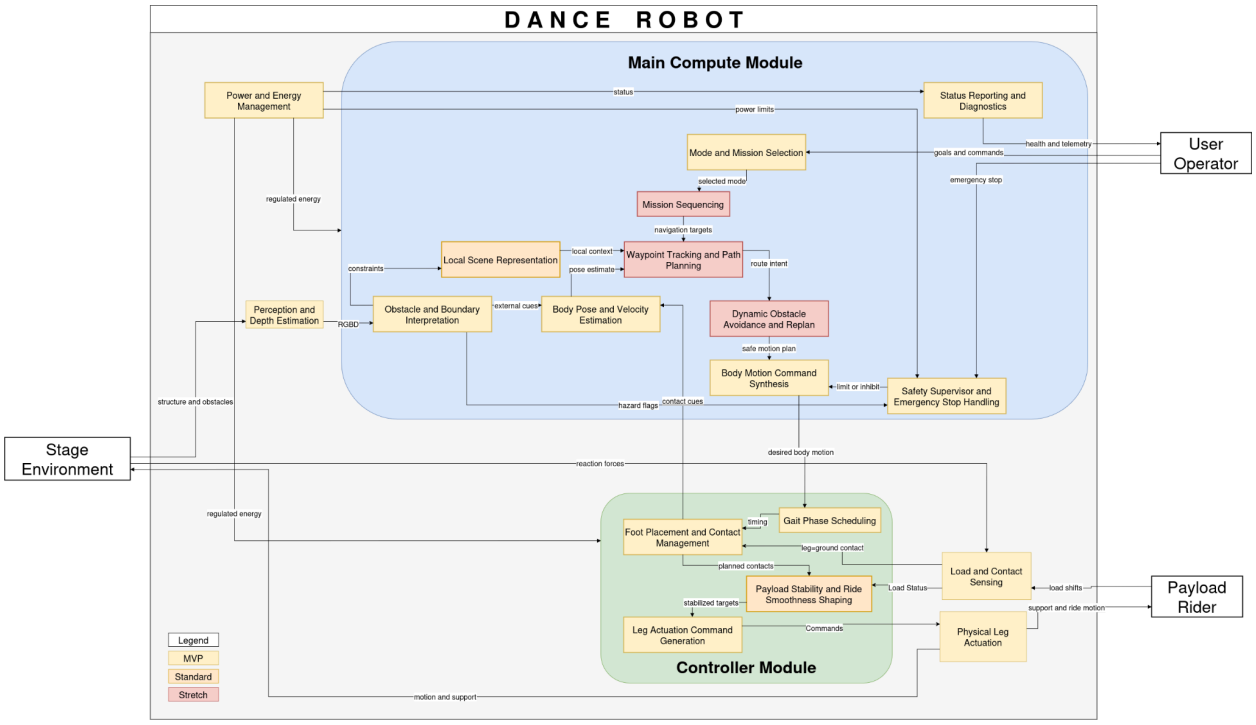

Figure 1. Functional architecture diagram.

The functional architecture decomposes the dance robot into a small set of purpose-specific functions that together enable safe, repeatable legged transport of a rider through an indoor space. At the top level, Mode and Mission Selection supports the allowed operating modes (pre-programmed motion, remote operator control, and user-driven joystick control), and provides a single place to arbitrate goals and permissions before any motion is commanded. A lightweight Status Reporting and Diagnostics function closes the loop for development and testing by exposing internal state, safety flags, and motion intent to the operator interface; however, the long-term intent is that the system remains fully embedded and portable, avoiding dependence on an external laptop during normal operation.

In the mobility pipeline, Body Pose and Velocity Estimation produces the robot-centric state needed to command controlled motion, while Obstacle and Boundary Interpretation provides environmental constraints (e.g., boundaries and hazards) used to limit motion near people and stage objects. These signals feed Body Motion Command Synthesis and the locomotion stack (Gait Phase Scheduling, Foot Placement and Contact Management, and Leg Actuation Command Generation), which together convert high-level motion intent into stable stance/swing timing and leg-level commands. This chain is directly responsible for meeting the fundamental mobility requirements, including forward/backward travel and zero-radius turning in both directions.

The autonomy capability is introduced as a standard feature through Local Scene Representation and Waypoint Tracking and Path Planning, which support navigation to a designated point within the required tolerance and time budget while carrying a rider. The architecture treats Dynamic Obstacle Avoidance and Replan as a stretch feature: it improves robustness in the presence of moving people by enabling reactive re-planning, but it is not required for fundamental operation because a remote operator can instead command velocities and manage avoidance during early demonstrations. Safety Supervisor and Emergency Stop Handling is a cross-cutting function that can inhibit or limit motion when hazards are detected, ensuring safe operation for the rider and surrounding environment and providing explicit emergency stop capability. Finally, Power and Energy Management regulates energy distribution and enforces power limits in support of sustained operation, including the 30-minute battery-life requirement.

4. Design trade studies

This section evaluates the critical architectural decisions for the Team 2 Dance Robot, focusing on how selected mechanical and electronic components fulfill the unique requirements of the AXIS Dance Company performance—specifically the need for a 170 lb payload capacity, a $900 budget, and "species atypical" movement.

4.1 Locomotion and actuation concept trade study

The primary challenge is selecting a leg mechanism that balances mechanical simplicity with the stability required for an unstrapped rider. We compared the Klann Linkage against the Jansen Linkage (popularized by the Strandbeest) and high-DOF Articulated Robotic legs.

Table 1. Locomotion and actuation concept trade study.

Feature	**Klann Linkage**	**Jansen Linkage**	**Articulated (3-DOF)**
Complexity	Moderate (6 links per leg)	High (8-12 links per leg)	High (Software-intensive)
Actuators	1 motor can drive 3 legs	1 motor can drive 3 legs	3 motors per leg (18 total)
Control Ease	Simple: Gait is mechanically encoded	Simple: Gait is mechanically encoded	Difficult: Requires Inverse Kinematics
Payload	Excellent: Robust force transmission	Moderate: Many joints increase flex	Variable: Limited by motor torque
Step Height	High; good for obstacle clearance	Relatively low; "strandbeest" style	Fully adjustable via software
Aesthetics	biological "walk"	biological "walk"	Realistic, insectoid movement

We selected the Klann linkage over the Jansen linkage or articulated legs to meet the high-stakes demands of an AXIS Dance Company performance, prioritizing mechanical reliability, payload dynamics, and safety. By mechanically encoding the walking gait, the Klann linkage eliminates the "software trip" risks inherent in complex Inverse Kinematics, ensuring the rider's safety is hard-wired into the physics of the machine rather than dependent on a digital control loop. This 6-bar mechanism excels in vertical force transmission, efficiently distributing the 320 lb system load across structural links to protect motor gear teeth from excessive stress. Furthermore, the Klann design offers superior step clearance compared to the Jansen linkage, providing the reliability needed to navigate stage obstacles and taped courses without snagging.

4.2 Motor Selection Trade Study

The primary drive motors must provide enough torque to rotate the Klann crankshafts under a total system load of approximately 320 lbs (170 lb rider + 150 lb robot) while remaining under the $2500 budget and 70 dB noise limit. We compared three distinct motor architectures:

Table 2. Drive Motor Selection for Klann Crankshafts

Selection Criteria	**High-Torque DC Gear Motor**	**Smart Servo (e.g., Dynamixel)**	**NEMA 23 Stepper + Gearbox**
Torque Output	Excellent: High reduction ratios provide massive lifting force.	Moderate: High torque models are extremely expensive.	High: Great holding torque, but loses torque at speed.
Cost (per unit)	Low ($40–$80): Leaves more budget for structural aluminum.	High ($150–$300): Would exceed the $900 project limit.	Medium ($80–$120): Requires expensive high-current drivers.
Control Logic	Simple: PWM for speed and H-bridge for direction.	Complex: Requires specific TTL/RS485 bus communication.	Moderate: Requires pulse/direction step counting.
Acoustics	Quiet: Low-RPM operation helps meet <70 dB requirement.	Variable: High-frequency PWM whine can be noisy.	Noisy: Distinctive "stepping" vibration and resonance.
Feedback	External: Requires separate encoders for 6" precision.	Internal: Built-in high-resolution encoders.	Open-Loop: Risk of losing steps under heavy payload.

We selected High-Torque DC Gear Motors for their superior performance-to-cost ratio, specifically tailored to the Klann linkage's mechanical demands. High reduction ratios provide the necessary "force multiplication" to prevent stalling during the high-load stance phase, while their cost-effectiveness allows budget reallocation toward the frame durability required for the 25 lb impact test. Furthermore, these motors support the 30-minute battery life requirement through better efficiency than stepper motors and meet the 6-inch navigation tolerance when paired with encoders. Finally, by operating at low RPMs to maintain a 1–4 mph walking speed, the motors stay well below the 70 dB threshold, ensuring acoustic compliance for the AXIS Dance Company performance.

4.3 Compute and Sensing Trade Study

The selection of the primary compute node and exteroceptive sensors is governed by the need for real-time obstacle detection and precise navigation. Given the requirement for autonomous point-to-point travel and a 30-minute battery life, the system must balance high computational throughput for image processing against power efficiency.

Table 3. Comparison of low-cost computers appropriate for embedded projects

Feature	**Raspberry Pi 4**	**Raspberry Pi 5**	**Nvidia Jetson Nano**
Processor (CPU)	Broadcom BCM2711 (Quad-core)	Broadcom BCM2712 (Quad-core)	Quad-core ARM A57
GPU/AI Acceleration	No dedicated hardware	No dedicated hardware	128-core Maxwell (CUDA)
RAM	8GB LPDDR4	8GB LPDDR4X	4GB LPDDR4
ROS2 Compatibility	Excellent (Ubuntu 22.04)	Excellent	Fair (Supports older versions)
Power Draw	5-8W	12-15W	5-10W

We have selected the Nvidia Jetson Nano. While the Raspberry Pi 4 and 5 offer competitive general-purpose computing and lower price points, the Jetson Nano's CUDA cores are essential for processing depth-map point clouds. To ensure rider safety in a dynamic dance environment, we require low-latency object detection. The Jetson Nano's dedicated GPU allows us to run computer vision algorithms more efficiently while maintaining a sustainable power draw for the 30-minute battery life requirement.

5. Cyberphysical architecture

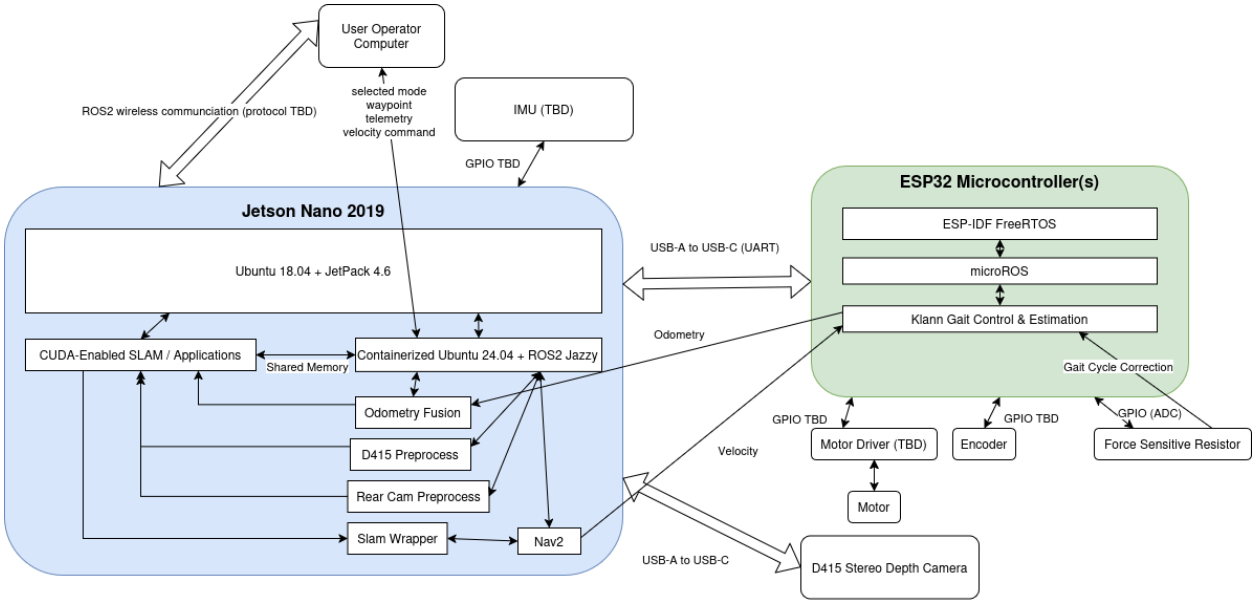

Figure 2. Cyberphysical architecture diagram.

The cyberphysical architecture partitions the robot into a high-level autonomy computer and a set of low-level real-time controllers. An embedded compute node (Jetson Nano) runs the perception and navigation stack in ROS2, while one or more ESP32 microcontrollers execute deterministic gait control, actuator interfacing, and local estimation under an RTOS. This split keeps the “heavy” workloads (camera processing, mapping/localization, and planning) on the Jetson, while timing-critical motor/encoder/foot-contact I/O stays on the ESP32 side.

In operation, the user/operator computer exchanges ROS2 messages wirelessly with the Jetson to set the active mode and provide mission inputs (e.g., waypoint commands or direct velocity commands consistent with allowable control modes). Onboard sensing enters through the depth camera (RealSense D415), optional rear camera, and IMU. The Jetson performs sensor preprocessing (camera pipelines, IMU ingestion), fuses proprioceptive and perception-derived estimates into a body pose/odometry state, and feeds that state to the navigation layer (SLAM wrapper and Nav2). The navigation layer produces body-level motion intent (velocity commands) that are sent over a UART link (USB-A to USB-C) to the ESP32 microcontroller(s). The ESP32 executes the gait controller and closes the low-level loops using encoder feedback and foot-contact proxies (e.g., force-sensitive resistor), producing motor-driver commands and applying gait-cycle correction. Low-level odometry and gait state are returned to the Jetson for fusion and for telemetry back to the operator station. Safety is treated as a first-class cross-cutting path: emergency stop controls are required, and hazard/power-limit conditions are expected to be able to inhibit or clamp commanded motion at the compute and/or microcontroller layers.

6. System description/depiction

6.1 Overall system overview

Figure 3. System depiction – overview.

The system consists of four primary subsystems: the main base structure, the wheel support assembly, one set of side legs and two sets of Klann-linkage-style legs. The main base structure provides the primary load-bearing platform and supports the seated performer. It serves as the structural backbone of the system and integrates all major mechanical components.

Beneath the main base, an electronics compartment is integrated between the main base and the wheel support assembly. This enclosed space houses the power distribution system, motor drivers, microcontrollers, wiring harnesses, and sensor modules. Locating the electronics centrally lowers the center of gravity, improves mass distribution, and protects sensitive components from external impact and debris.

The wheel support assembly is designed to carry the majority of the system weight. It provides static stability and reduces the load transmitted to the leg mechanisms during operation, thereby improving overall efficiency and durability.

The Four Klann-linkage leg assemblies are responsible for generating forward and backward locomotion. In addition, a pair of side-mounted legs driven by linear actuators are dedicated to lateral motion.

6.2 Klann linkage locomotion subsystem

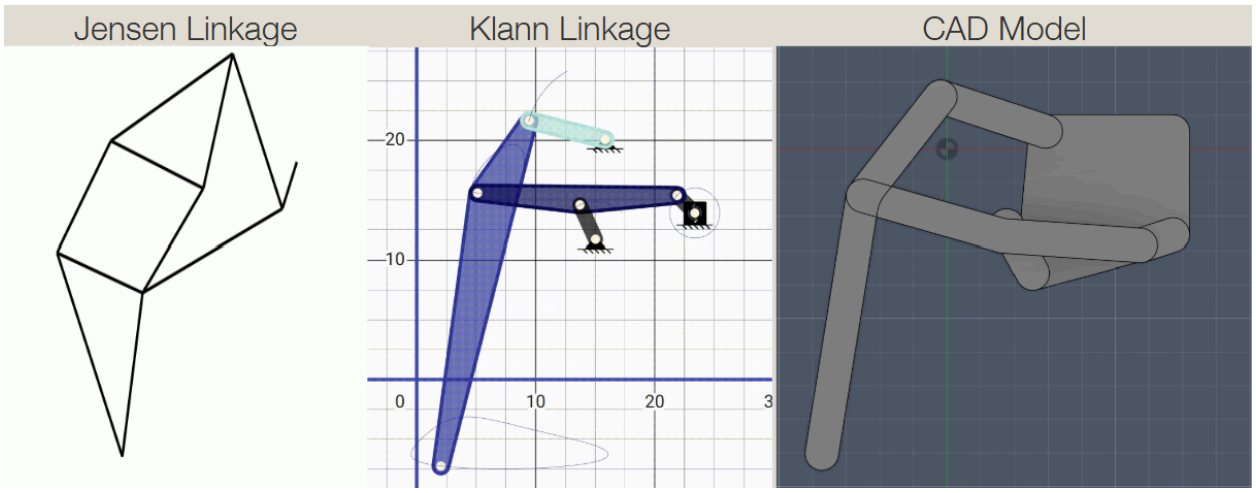

Figure 4. Klann design process

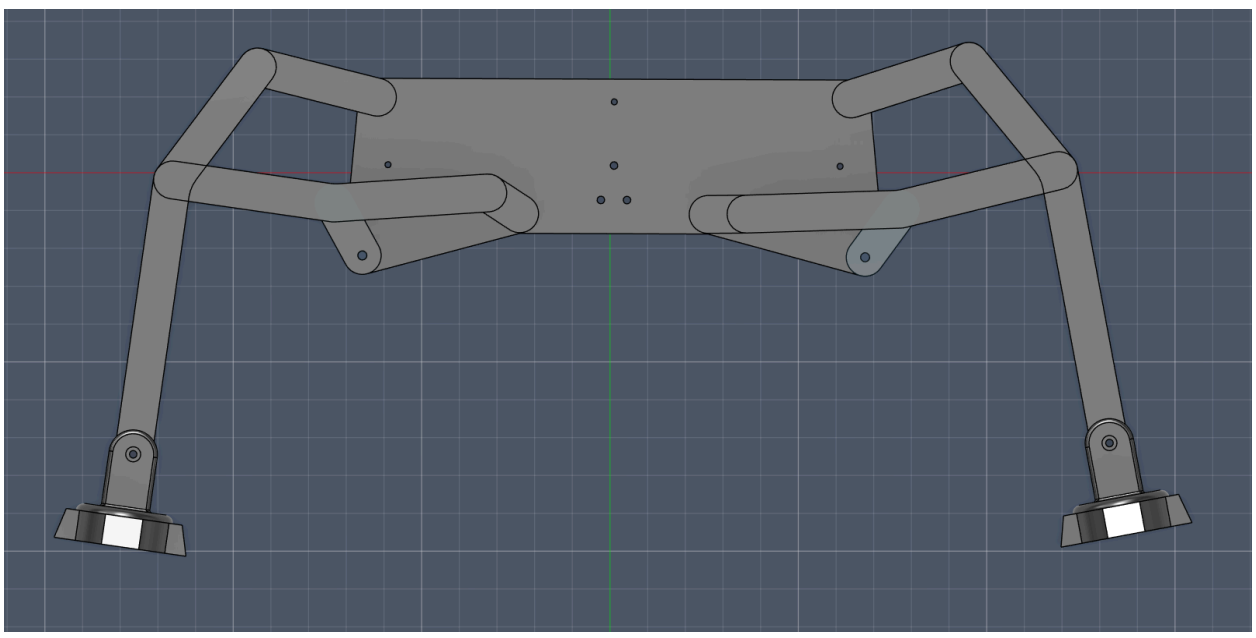

Figure 5. System depiction – Klann linkage.

To minimize the total number of actuators while maintaining effective walking performance, the team evaluated multiple linkage mechanisms capable of converting rotary motion into a walking gait. After comparative analysis, the Klann linkage was selected as the primary leg mechanism. Compared to the Jansen linkage, the Klann mechanism consists of fewer

links and joints, resulting in a mechanically simpler architecture with reduced assembly complexity, fewer tolerance stack-up concerns, and lower maintenance requirements.

In the current design, rotary DC gear motors drive the Klann crank shafts. Each crank shaft is supported by appropriate shaft support hardware, including bearings, shoulder bolts, linkage plates, pivot joints, and foot pads, as outlined in the mechanical budget. One motor is used per side of the system, driving the corresponding set of leg linkages.

These leg assemblies convert continuous rotary input into a cyclic walking trajectory, producing stable ground contact and forward propulsion. The synchronized linkage motion ensures consistent footfall patterns and load distribution during locomotion.

Turning is implemented using a differential-drive strategy similar to that of a tracked vehicle. By independently controlling the rotational speeds of the left and right motor-driven leg sets, the system can achieve zero-radius (0 radian) turning about its vertical axis.

6.3 Side leg subsystem

The side legs provide active stabilization, pose control, and lateral (side-to-side) body translation during locomotion and expressive motion. Specifically, they (1) increase the support polygon to reduce tip risk under payload, (2) help maintain rider height in the target range (2–3 ft), (3) enable controlled body roll/pitch for choreographed movements, and (4) provide controlled lateral shifting to reposition the body relative to the stance legs—supporting balance recovery and expressive “side glide” motions while staying within the 5’×5’ footprint requirement.

Each side leg is an articulated link assembly with an end foot pad, driven by a compact linear actuator (motor-internal) through a pinned joint/clevis interface. The CAD shows a multi-link arm with cutouts for weight reduction and mounting interfaces to the main frame.

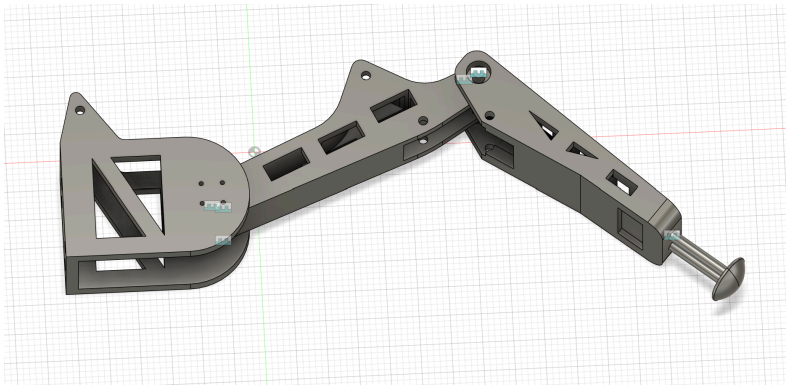

Figure 6. System depiction – side legs.

We baseline linear actuators (motor inside) for side-leg actuation due to relatively low integration complexity, precise position control, and reduced safety concerns compared with fluid power. Our provisional mechanical budget already includes two linear actuators with

100–200 mm stroke and 300–1000 N force range, along with clevis/rod ends and actuator mounting brackets.

However, due to expected cost, integration complexity, and stage-safety considerations, the current direction is shifting toward a motor-driven mechanism to achieve the required stabilization and lateral-shift motions with simpler integration into the overall electromechanical system.

6.4 Sensing and compute subsystem

The sensing and compute subsystem supplies basic environment awareness and motion state estimates needed for safe operation and (when enabled) waypoint navigation. It uses a forward stereo depth camera as the primary exteroceptive sensor, with an auxiliary rear-facing camera path for backing/turning awareness and an IMU for short-term motion stabilization and disturbance detection. Onboard compute handles sensor preprocessing and publishes perception/state outputs to the autonomy and safety layers, supporting teleoperation during early testing and autonomous navigation in later phases.

7. Project management

7.1 Schedule

Table 4. Schedule

	W5	W6	W7	W8	W9	W10	W11	W12	W13	W14	W15
CAD Design											
Simulation											
Locomotion Control Implementation											
Assembly											
Circuit Assembly											
Localization Implementation											
Full Assembly Testing											
Computer system setup											
Computer system testing											

Table 4 shows a high-level, abbreviated form of our plan for the remainder of the semester with regards to every major component of our robot prototype. Please refer to our github organization and project (found in team website) for per-task scheduling.

7.2 Team member responsibilities

Table 5. Member responsibilities.

Team Member	Primary Responsibilities	Secondary Responsibilities
Daniel Gonzalez Lazaro	● Systems Integration ● Microcontroller programming ● ROS2 programming ● Github management ● Website updating ● SLAM implementation ● Power circuit ● Sensor and computer circuit	● Locomotion control implementation ● Simulation setup and troubleshooting
Yechen Zheng	● Mechanism Design ● Parts manufacturing ● Robot Fabrication and Assembly	● Control Optimization ● Gait simulation
Hongfei Liu	● Assembly ● Locomotion control optimization ● Parts manufacturing ● Robot Fabrication and Assembly	
Hongru Liu	● Assembly ● Parts tracking and ordering ● Parts manufacturing ● Robot Fabrication and Assembly	

7.3 Budget

Table 6. Allocation of $2500 Motor Stipend

Item	Description / Notes	Qty	Unit Cost (USD)
DC gear motors (Klann drive)	Rotary motors	2–4	300-500

Table 7. Estimated budget.

Motor mounting brackets	Motor-to-frame mounts	2–4	15–50
Shaft couplers	Motor-to-crank shaft couplers	2–4	10–30
Crank shafts	Steel or aluminum shafts	2–4	15–50

Pillow block / radial bearings	Support crank shafts	4–8	10–40
Shaft collars	Axial shaft retention	6–12	5–15
Klann linkage plates	Aluminum / G10 link sets	4 sets	25–120
Shoulder bolts / joint pins	Klann pivot joints (M5–M8)	4 sets	15–60
Bearings / bushings	Flanged bearings or bronze bushings	4 sets	20–80
Spacers & washers	Aluminum / nylon spacers	4 sets	5–20
Klann foot pads	Rubber / TPU feet	4	5–25
Klann leg mounting brackets	Leg-to-frame interface	4	10–50
Linear actuators (motor inside)	100–200 mm stroke, 300–1000 N	2	40–250
Clevis / rod ends	Actuator end joints	4	10–40
Actuator mounting brackets	Frame and leg mounts	2 sets	20–60
Actuator leg structures	Links / tubes for actuator legs	2 sets	20–60
Actuator leg bearings	Pivot bearings / bushings	2 sets	10–30
Actuator leg foot pads	Rubber / TPU feet	2	5–25
Main frame extrusions / plates	6061 aluminum	1 set	150–600
Side plates / hubs	Thick leg-mount plates	2–4	40–200
Cross bracing / gussets	Structural stiffening	1 set	30–150
Standoffs / spacers	Frame spacing	1 set	20–100
Fastener assortment	M3–M8 bolts, nuts, washers	1 lot	50–200
Thread inserts	Heat-set or press-fit	1 lot	10–60
Adhesives & consumables	Loctite, epoxy	1 lot	10–40
Battery	Powering systems	1	150
Office chair		1	80
Fabrication allowance	Extra material / mistakes		10–25%

Thus the estimated total, excluding motors is in the range of $670–$1300

7.4 Risk management

We have identified the following potential issues in constructing our prototype:

Side leg size/geometry not finalized: If the side-leg length or placement is wrong, it can cause mechanical interference, reduce stability under payload, or violate footprint/foldability constraints. Mitigation: define a fixed kinematic/packaging envelope early, run quick CAD interference checks across worst-case poses (extended, retracted, turning, folded), and freeze geometry before releasing parts.

Actuation selection uncertainty (hydraulic / pneumatic / motor / linear actuator): The side-leg actuator choice drives force capacity, controllability, noise, safety, and integration; a poor choice could yield inadequate performance or unsafe behavior. Hydraulics are likely too costly/complex and pneumatics may not hold position reliably, so the current direction is a motor-driven solution. Mitigation: set minimum force/stroke/speed/noise targets, bench-test a short list of candidates, and keep mounts modular so we can swap actuators without redesigning the chassis.

Materials and joint hardware not selected: Poor material/joint choices can lead to buckling, backlash, and wear, reducing stability and safety. Mitigation: standardize pivots using shoulder bolts/joint pins with bearings/bushings, follow the planned structural material approach, and do quick worst-load strength sanity checks before fabrication.

Late design decisions delay ordering and integration: Delays in freezing side-leg geometry/actuation can push back CAD release, procurement, assembly, and system testing. Mitigation: enforce a design-freeze milestone, order long-lead/common components early, and integrate in stages so base locomotion and controls testing can proceed in parallel.

Budget creep from actuator/frame changes and rework: Upgrades, custom machining, and iteration can exceed the mechanical budget. Mitigation: maintain a live budget with contingency, favor off-the-shelf parts and simple fabrication, and prototype high-uncertainty subsystems cheaply before committing to final hardware.

8. References

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